

Re-analysis Summary Report for Peer-evaluation

Paper ID: Christensen_EurJournPersonality_2018_8R9d

Re-analyst's ID: ql2m7

Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

For both tasks, you should provide your evaluation via this survey: <https://forms.gle/EAREaC6JjN2PWFfR6>

The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:

... the high openness to experience group's network was more interconnected ... than the low openness to experience group (p. 480.)

The corresponding article

Here you can find the original article: https://osf.io/6ve3b/?view_only=c874fcf097644e82a320679d4641163a and the original data: https://osf.io/6zh34/?view_only=3e5bfed2e1f041dbbb4376e9c0c5f70b

They are provided only as context of the re-analysis.

In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:

The statistical analysis included several consecutive steps. First, we had to divide the sample into two contrasting groups based on the level of openness to experience. However, all participants had several scores on openness to experience because in the original study they were asked to complete two or three measures simultaneously. The measures were NEO-PI-3, NEO-FFI-3, and BFAS. Most participants completed only NEO-FFI-3 and BFAS ($n = 379$; referred to as "samples 1-3") while others completed all three measures ($n = 137$; referred to as "sample 2"). I conducted a correlation analysis using Pearson's correlation coefficient that revealed a high inter-relationship between the results of three measures ($r_s = .63-.89$). Next, I proceeded with the exploratory factor analysis (EFA) in order to extract estimated factor scores for openness to experience as a latent variable. The EFA was performed separately for "sample 1-3" and "sample 2" based on the matrix of Pearson's correlation coefficients using principal axis factoring (PAF). For each sample estimated factor scores were obtained following Bartlett's method. After joining factor scores across samples, we calculated the median of factor scores' distribution to use it further in forming comparison groups. The choice of median as a threshold was justified by the need to have an equivalent or a comparable number of participants in each group to preserve the maximum statistical power in testing networks' structural parameters (see details

below). Thus, I divided the whole sample ($N = 516$) into groups of “low” (factor score $<$ median) and “high” (factor score $\geq$ median) openness to experience. Second, I preprocessed the data on the verbal fluency (VF) task in order to proceed with the network analysis allowing to examine various macro-level network parameters reflecting the structural architecture of semantic memory. Although participants could generate up to 50 responses, close examination of missing values uncovered that the maximum number of responses equaled 35. Thus, we eliminated all columns with 100% of missing values ($m = 15$) and proceeded further with the remaining columns. The next logical step was an application of text-cleaning algorithms compiled in such R packages as “SemNetCleaner” and “SemNetDictionaries”, which auto-corrected all potential errors and typos based on the animals’ dictionary (i.e., a collection of more than 1000 nouns designating various animals) and converted VF responses into binary matrices of words’ co-occurrences. After that, we finalized matrices so that each VF response has been given by at least two participants, equated responses across the networks for two groups (i.e., “low” and “high” openness), estimated words similarity using Pearson’s correlation, estimated both networks with Triangulated Maximally Filtered Graph (TMFG; the only filtering method implemented in “SemNet” package for correlation-based networks), and plotted them against each other to visually inspect their difference. Lastly, we conducted statistical computations based on two networks. The preliminary step was to define the most appropriate networks’ properties to establish the level of inter-connectedness between semantic units because it was the main aim of the initial research claim. Taking into account previous studies on the relationship between the structure of semantic memory and various creativity facets, we finished with three macro-level network parameters: (a) average shortest path length (ASPL; an average shortest number of edges needed to get between any pair of nodes), (b) clustering coefficient (CC; the extent that two neighbors of a node will be neighbors themselves), and (c) maximum modularity coefficient (Q; how well networks partitions in smaller networks). After computing all three parameters for “low” and “high” openness groups, we run a test to confirm that the obtained VF networks’ parameters corresponded to the Watts-Strogatz model (“small-world model”) rather than to the Erdős-Rényi model (“random network”). The latter suggestion was successfully established using Viger and Latapy’s Markov chain Monte Carlo (MCMC) algorithm with 1000 iterations (all p-values $< .001$) and provided an opportunity to continue with the statistical comparison of networks’ parameters between “low” and “high” openness groups. For the latter purpose, we utilized a case-wise bootstrap approach with 1000 iterations and compared the obtained sampling distributions for ASPL, CC, and Q using a t-test for independent samples. Results showed that lower level of openness to experience corresponded to higher average distances among nodes (ASPL; $t(1998) = 19.60$, $p < .001$), lower clusterization (CC; $t(1998) = -15.98$, $p < .001$), and higher modularity (Q; $t(1998) = 12.71$, $p < .001$). In sum, our analysis corroborated the claim provided in the original publication: The group with high openness to experience compared to the group with low openness to experience had a better-connected structure of semantic memory.

The analyst’s conclusion: The group with high openness to experience compared to the group with low openness to experience had better-connected structure of semantic memory

The analyst’s categorization of the result:

The results show evidence for the relationship/effect as described in the claim provided in your task

In Task 2, the analysts received the following instruction to conduct the analysis:

In the partial network that you use, the rate of nodes remaining should be 90%.

Here, the analyst reported the following steps and results of the analysis:

The statistical analysis included several consecutive steps. First, we had to divide the sample into two contrasting groups based on the level of openness to experience. However, all participants had several scores on openness to experience because in the original study they were asked to complete two or three measures simultaneously. The measures were NEO-PI-3, NEO-FFI-3, and BFAS. Most participants completed only NEO-FFI-3 and BFAS ($n = 379$; referred to as “samples 1–3”) while others completed all three measures ($n = 137$; referred to as “sample 2”). I conducted a correlation analysis using Pearson’s correlation coefficient that revealed a high inter-relationship between the results of three measures ($r_s = .63-.89$). Next, I proceeded with the exploratory factor analysis (EFA) in order to extract estimated factor scores for openness to experience as a latent variable. The EFA was performed separately for “sample 1–3” and “sample 2” based on the matrix

of Pearson’s correlation coefficients using principal axis factoring (PAF). For each sample estimated factor scores were obtained following Bartlett’s method. After joining factor scores across samples, we calculated the median of factor scores’ distribution to use it further in forming comparison groups. The choice of median as a threshold was justified by the need to have an equivalent or a comparable number of participants in each group to preserve the maximum statistical power in testing networks’ structural parameters (see details below). Thus, I divided the whole sample ($N = 516$) into groups of “low” (factor score < median) and “high” (factor score $\geq$ median) openness to experience. Second, I preprocessed the data on the verbal fluency (VF) task in order to proceed with the network analysis allowing to examine macro-level network parameters reflecting the structural architecture of semantic memory. Although participants could generate up to 50 responses, close examination of missing values uncovered that the maximum number of responses equaled 35. Thus, we eliminated all columns with 100% of missing values ($m = 15$) and proceeded further with the remaining columns. The next logical step was an application of text-cleaning algorithms compiled in such R packages as “SemNetCleaner” and “SemNetDictionaries”, which auto-corrected all potential errors and typos based on the animals’ dictionary (i.e., a collection of more than 1000 nouns designating various animals) and converted VF responses into binary matrices. After that, we finalized matrices so that each VF response has been given by at least two participants, equated responses across the networks for two groups (i.e., “low” and “high” openness), estimated words similarity using Pearson’s correlation, estimated both networks with Triangulated Maximally Filtered Graph (TMFG; the only filtering method implemented in “SemNeT” package for correlation-based networks), and plotted them against each other to visually inspect their difference. Lastly, we conducted statistical computations based on two networks. The preliminary step was to define the most appropriate networks’ property to establish the level of inter-connectedness between semantic units because it was the main aim of the initial research claim. Taking into account previous studies on the relationship between the structure of semantic memory and various creativity facets, we finished with three macro-level network parameters as potential candidates: (a) average shortest path length (ASPL; an average shortest number of edges needed to get between any pair of nodes), (b) clustering coefficient (CC; the extent that two neighbors of a node will be neighbors themselves), and (c) maximum modularity coefficient (Q; how well networks partitions in smaller networks). Since the hypothesis focused on the interconnection between nodes in two groups and only one result of statistical analysis was allowed, we limited the scope of our analysis to the ASPL. The choice of ASPL was predetermined by the fact that the corresponding indicator reflects the average shortest number of edges needed to move between a pair of nodes. Thus, the shorter average distances imply tighter connections among nodes in the network. In addition, according to the task instruction, the number of preserved nodes in the partial networks was fixed to 90%. After computing ASPL parameter for “low” and “high” openness groups, we run a test to confirm that the obtained VF networks’ parameter corresponded to the Watts-Strogatz model (“small-world model”) rather than to the Erdős-Rényi model (“random network”). The latter suggestion was successfully established using Viger and Latapy’s Markov chain Monte Carlo (MCMC) algorithm with 1000 iterations (all p-values < .001) and provided an opportunity to continue with the statistical comparison of networks’ parameter between “low” and “high” openness groups. For the latter purpose, we utilized a node-wise bootstrap approach with 1000 iterations and compared the obtained sampling distributions for ASPL using a t-test for independent samples. Results showed that lower level of openness to experience corresponded to higher average distances among nodes (ASPL; $t(1998) = 57.13$, $p < .001$, $d = 2.56$). In sum, our analysis corroborated the claim provided in the original publication: The group with high openness to experience compared to the group with low openness to experience had a better-connected structure of semantic memory.

Analysis language/software/code (optional to check):

Task 1: R

Task 2: R

Analysis codes: <https://osf.io/f7prm/>